Experiment 13
Sampling Theorem Verification and Sampling-Frequency Selection

```
clc;
clear;
close all;

%% 1. Original Signal
fm = 50;
T = 0.06;
dt = 1e-5;

t = 0:dt:T;
x = cos(2*pi*fm*t);

%% 2. Sampling at Different Rates
fs_list = [0.5*fm, fm, 2*fm, 5*fm, 10*fm];

labels = {
    'fs = 0.5 fm (Aliased)'
    'fs = fm (Undersampled)'
    'fs = 2 fm (Nyquist Rate)'
    'fs = 5 fm (Oversampled)'
    'fs = 10 fm (Oversampled)'
    };

figure('Name', 'Sampling at Different Rates');

for k = 1:length(fs_list)

    fs = fs_list(k);

    n = 0:floor(T*fs);
    ts = n/fs;

    xn = cos(2*pi*fm*ts);

    subplot(5,1,k);
    plot(t, x, 'b-', 'LineWidth', 1);
    hold on;
    stem(ts, xn, 'r', 'filled');
    title(labels{k});
    xlabel('Time (s)');
    ylabel('Amplitude');
    grid on;
    hold off;

end

%% 3. Sinc Reconstruction and Error Calculation
fprintf('Nyquist rate = %.2f Hz\n', 2*fm);

err = zeros(size(fs_list));

for k = 1:length(fs_list)

    fs = fs_list(k);

    n = 0:floor(T*fs);
    ts = n/fs;

    xn = cos(2*pi*fm*ts);

    % Vectorized sinc interpolation
    tau = t(:) - ts;
    x_recon = sinc(fs*tau) * xn(:);
    x_recon = x_recon.';

    err(k) = mean(abs(x - x_recon));

    fprintf('fs = %6.1f Hz, fs/fm = %4.1f, Error = %.5f\n', ...
        fs, fs/fm, err(k));

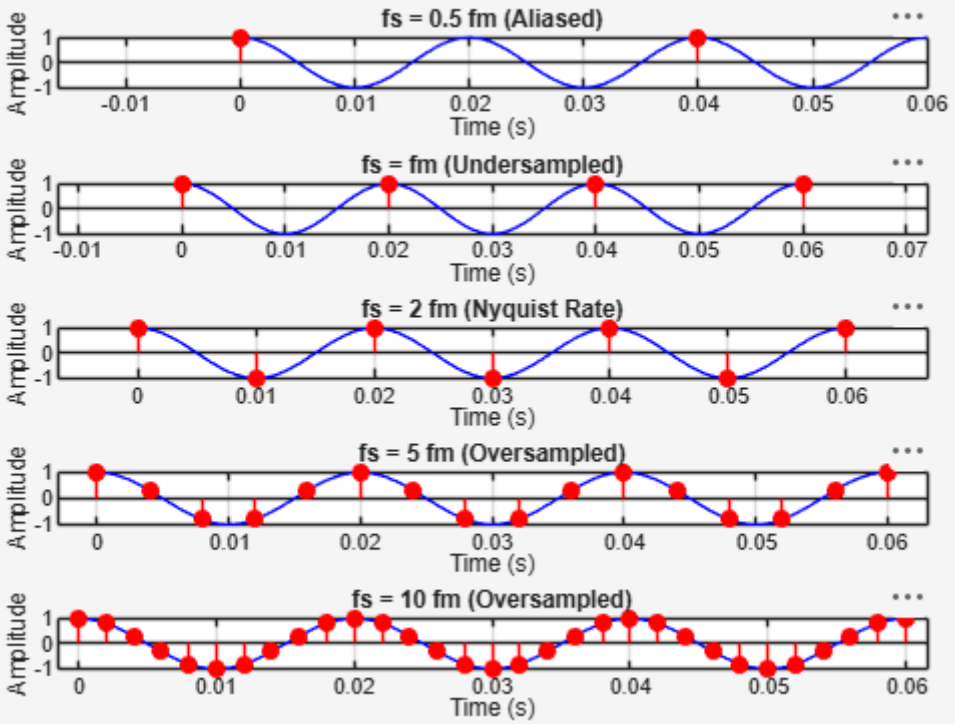
end

%% 4. Reconstruction at 10 Times the Message Frequency
fs = 10*fm;

n = 0:floor(T*fs);
ts = n/fs;
```

Code

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Output